



- 3 Two different types of juice extractor, machine X and machine Y , are being compared. The manufacturer claims that machine Y extracts more juice per orange on average than machine X .

A random sample of 20 oranges is selected and each carefully measured. Oranges of a similar size and mass are then paired and numbered from 1 to 10. One orange from each pair is randomly allocated to machine X with the other allocated to machine Y . The amount of juice, in ml, extracted from each orange is recorded in the following table.

	pair									
	1	2	3	4	5	6	7	8	9	10
machine X	65	73	58	61	72	79	64	65	69	71
machine Y	68	72	64	63	75	82	63	63	72	74

- (a) Use a t -test to test the manufacturer's claim at the 1% significance level. [7]

[illegible]



(b) State an assumption required for the test in **3(a)** to be valid.

[1]

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The manufacturer notices that the amount of juice extracted from the oranges in pair 4 were recorded incorrectly. In fact, the amount of juice extracted from machine X was 60ml and from machine Y was 62 ml.

(c) Explain, with justification, whether the conclusion of the test in 3(a) remains the same.

[2]

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